Supplementary Material — Machine-learning models of hydrothermal biomass conversion trained on literature data: transfer to unseen studies and study-calibrated prediction intervals

S1. Literature screening

PubMed was queried on 2026-11-10 for machine-learning studies of hydrothermal liquefaction or carbonization published 2015–2026 (query string in `supplementary/literature/pubmed_corpus.py`). Titles and abstracts of the 44 primary studies were screened with pre-defined regular expressions for the validation design, uncertainty quantification and physical constraints. Only 2 abstracts mentioned grouped, study-level or external validation and 4 mentioned any form of uncertainty quantification. Abstract-level screening under-counts practices that appear only in the full text; the counts are therefore lower bounds.

Table S1. PubMed-indexed ML studies of hydrothermal conversion (abstract-level screening; n.s. = not stated in abstract).

Year	Journal	First author	DOI	Target(s)	Methods	Validation (abstract)	R^2 in abstract	UQ	Physics
2018	Bioresour Technol	Li	10.1016/j.biortech.2018.04.066	hydrochar yield/properties	RF	n.s.	n.s.	n.s.	n.s.
2019	Waste Manag	Ismail	10.1016/j.wasman.2018.12.044	carbon recovery	ANN/MLP	n.s.	n.s.	n.s.	n.s.
2021	Bioresour Technol	Zhang	10.1016/j.biortech.2021.126011	bio-oil/biocrude yield; N content/h	RF; GBR/GBM	n.s.	n.s.	n.s.	n.s.

Year	Journal	First author	DOI	Target(s)	Methods	Validation (abstract)	R^2 in abstract	UQ	Physics
				eterocycles					
2022	Bioresour Technol	Katongtung	10.1016/j.biortech.2021.126278	bio-oil/biocrude yield; HHV/energy	XGB; GBR/GBM	random/k-fold	n.s.	n.s.	n.s.
2022	Bioresour Technol	Zhou	10.1016/j.biortech.2021.126354	n.s.	RF	n.s.	n.s.	n.s.	n.s.
2022	Bioresour Technol	Zhou	10.1016/j.biortech.2022.126923	n.s.	SVR	n.s.	n.s.	n.s.	n.s.
2022	Bioresour Technol	Leng	10.1016/j.biortech.2022.127791	bio-oil/biocrude yield; N content/heterocycles	RF	n.s.	0.92;0.82	n.s.	n.s.
2022	Bioresour Technol	Gruber	10.1016/j.biortech.2022.128071	hydrochar yield/properties	ANN/MLP	n.s.	0.997;0.978	n.s.	n.s.
2022	Water Sci Technol	Divband Hafshejani	10.2166/wst.2022.264	hydrochar yield/properties; adsorption	ANN/MLP; SVR	n.s.	n.s.	n.s.	n.s.
2022	Open Res Eur	Haarlemmer	10.12688/openreseurope.14915.2	bio-oil/biocrude yield	n.s.	n.s.	n.s.	n.s.	n.s.
2023	Chemosphere	Yoo	10.1016/j.chemosphere.2022.137074	hydrochar yield/properties; adsorption	ANN/MLP	n.s.	n.s.	n.s.	n.s.
2023	Bioresour Technol	Zhu	10.1016/j.biortech.2022.128454	hydrochar yield/properties; HHV/energy; carbon recovery	n.s.	n.s.	0.983	n.s.	n.s.
2023	Bioresour Technol	Chen	10.1016/j.biortech.2023.128893	n.s.	RF; ANN/MLP; SVR	n.s.	n.s.	n.s.	n.s.
2023	Bioresour Technol	Zhao	10.1016/j.biortech.2023.129223	hydrochar yield/properties; adsorption	GBR/GBM; DT	n.s.	0.93	n.s.	n.s.
2023	Heliyon	Kohzadi	10.1016/j.heliyon.2023.e21258	hydrochar yield/properties; adsorption	ANN/MLP	n.s.	n.s.	n.s.	n.s.
2023	Commun Chem	Wang	10.1038/s42004-	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

Year	Journal	First author	DOI	Target(s)	Methods	Validation (abstract)	R^2 in abstract	UQ	Physics
			023-01077-z						
2024	Sci Total Environ	Ma	10.1016/j.scitotenv.2023.168279	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
2024	Bioresour Technol	Liu	10.1016/j.biortech.2023.130073	hydrochar yield/properties	RF; XGB; GBR/GBM; SVR	n.s.	n.s.	n.s.	n.s.
2024	Chemosphere	Vinayagam	10.1016/j.chemosphere.2024.141130	adsorption	ANN/MLP	n.s.	0.99	n.s.	n.s.
2024	Sci Total Environ	Liu	10.1016/j.scitotenv.2024.170779	HHV/energy; N content/heterocycles	RF; GBR/GBM	n.s.	n.s.	n.s.	n.s.
2024	Chemosphere	Mahmoud	10.1016/j.chemosphere.2024.141746	hydrochar yield/properties; adsorption	ANN/MLP	n.s.	0.989	n.s.	n.s.
2024	Sci Total Environ	Lahori	10.1016/j.scitotenv.2024.172810	hydrochar yield/properties	linear/ridge	n.s.	n.s.	n.s.	n.s.
2024	J Phys Chem A	Srinivasan	10.1021/acs.jpca.4c01592	n.s.	ANN/MLP	n.s.	n.s.	n.s.	n.s.
2024	Sci Total Environ	Zhang	10.1016/j.scitotenv.2024.173939	n.s.	GBR/GBM	n.s.	0.99;0.80; 0.89	n.s.	n.s.
2024	J Hazard Mater	Liu	10.1016/j.jhazmat.2024.135853	hydrochar yield/properties; adsorption	GBR/GBM	n.s.	n.s.	n.s.	n.s.
2024	Sci Total Environ	Tong	10.1016/j.scitotenv.2024.176780	hydrochar yield/properties	RF; GBR/GBM	n.s.	0.86	n.s.	n.s.
2024	F1000Res	Katongtung	10.12688/f1000research.156514.3	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
2025	Environ Res	Xiong	10.1016/j.envres.2025.122168	hydrochar yield/properties	RF	n.s.	n.s.	n.s.	n.s.
2025	Waste Manag	Biswas	10.1016/j.wasman.2025.115054	hydrochar yield/properties	ANN/MLP	random/k-fold	n.s.	n.s.	n.s.
2025	Bioresour Bioprocess	Hou	10.1186/s40643-025-00979-1	hydrochar yield/properties; HHV/energy	XGB; GBR/GBM; ANN/MLP; DT;	n.s.	0.98;0.94	yes	n.s.

Year	Journal	First author	DOI	Target(s)	Methods	Validation (abstract)	R^2 in abstract	UQ	Physics
					linear/ridge				
2026	Bioresour Technol	Igwegbe	10.1016/j.biortech.2025.133282	hydrochar yield/properties; HHV/energy; carbon recovery	XGB; SVR; linear/ridge	n.s.	0.87;0.938;0.85	n.s.	n.s.
2026	Bioresour Technol	Yan	10.1016/j.biortech.2025.133451	hydrochar yield/properties; carbon recovery	n.s.	n.s.	n.s.	n.s.	n.s.
2026	Bioresour Technol	Liu	10.1016/j.biortech.2026.133921	hydrochar yield/properties; N content/heterocycles	RF	grouped/external	0.882;0.877;0.894	n.s.	n.s.
2026	Bioresour Technol	Lu	10.1016/j.biortech.2026.134279	N content/heterocycles	GBR/GBM	n.s.	n.s.	n.s.	n.s.
2026	Bioresour Technol	Gao	10.1016/j.biortech.2026.134360	n.s.	RF; GBR/GBM	n.s.	0.82;0.79;0.65	n.s.	n.s.
2026	Environ Sci Technol	Guo	10.1021/acs.est.6c01833	hydrochar yield/properties; HHV/energy	n.s.	n.s.	n.s.	n.s.	n.s.
2026	RSC Adv	Do	10.1039/d6ra01824f	adsorption	ANN/MLP	n.s.	n.s.	n.s.	n.s.
2026	Bioresour Technol	Khan	10.1016/j.biortech.2026.134909	hydrochar yield/properties	n.s.	random/k-fold	0.85	yes	n.s.
2026	Bioresour Technol	Li	10.1016/j.biortech.2026.135052	hydrochar yield/properties	linear/ridge	n.s.	n.s.	n.s.	n.s.
2026	RSC Adv	Asim	10.1039/d6ra05073e	adsorption	n.s.	n.s.	n.s.	n.s.	n.s.
2026	J Environ Manage	Ma	10.1016/j.jenvman.2026.130813	hydrochar yield/properties; HHV/energy	GBR/GBM	n.s.	0.890;0.831	yes	n.s.
2026	Bioresour Technol	Huang	10.1016/j.biortech.2026.136066	hydrochar yield/properties; adsorption	n.s.	n.s.	n.s.	n.s.	yes
2026	Environ Sci Technol	Liu	10.1021/acs.est.6c05758	HHV/energy	n.s.	n.s.	n.s.	n.s.	n.s.
2027	Bioresour	Elfetni	10.1016/	HHV/	XGB	grouped/	0.844	yes	yes

Year	Journal	First author	DOI	Target(s)	Methods	Validation (abstract)	R^2 in abstract	UQ	Physics
	Technol		j.biortech.2026.135587	energy		externalrandom/k-fold			

S2. Pre-registered protocol and deviations

The analysis protocol (`code/config/PROTOCOL.md`) was committed to version control before any model was fitted, together with the locked list of 38 test studies. Deviations, each logged with date and reason: (i) the tabular foundation model could not be run because its weights require a personal licence acceptance; (ii) clarification that every learner was tuned separately for each feature set; (iii) the ridge-penalty grid of the meta-regression was widened after its selected value hit the registered upper bound, and three univariate study-level tests were added as secondary analyses; (iv) post-hoc review analyses E9 (base learners), E10 (cross-study duplicate audit) and E11 (overhead), and a post-hoc sensitivity analysis of the locked test excluding the single study with the largest squared-error share; (v) an exploratory, post-hoc locally adaptive variant of the study-calibrated intervals (SC-N, Section S4) was added after pipeline verification showed that constant-width study-calibrated intervals are wide. GPBoost was run as registered; its study random intercept absorbed most of the variance because feedstock descriptors are nearly constant within a study, so its out-of-study predictions reverted towards a nearly flat fixed-effect function.

S3. Additional benchmark and sensitivity results

Table S2. Benchmark with base + protocol (P) features.

Target	Learner	Random CV RMSE	Grouped CV RMSE	Grouped CV R^2
Bio-oil	LightGBM	7.53	16.01	0.209
Bio-oil	Random forest	8.33	16.23	0.188
Bio-oil	XGBoost	7.65	16.33	0.178

Target	Learner	Random CV RMSE	Grouped CV RMSE	Grouped CV R^2
Bio-oil	Extra trees	8.33	16.35	0.176
Bio-oil	SVR	11.04	16.45	0.165
Bio-oil	CatBoost	7.66	16.62	0.148
Bio-oil	HistGB	7.81	16.9	0.119
Bio-oil	k-NN	10.01	17.15	0.093
Bio-oil	GPBoost	17.29	17.61	0.044
Bio-oil	Ridge	14.15	17.73	0.03
Bio-oil	Mean	18.02	18.12	-0.013
Bio-oil	MLP	8.63	23.32	-0.681
Solid	LightGBM	7.03	13.65	0.544
Solid	XGBoost	7.06	14.08	0.516
Solid	HistGB	7.07	14.17	0.51
Solid	Random forest	8.6	14.22	0.506
Solid	CatBoost	7.66	14.64	0.476
Solid	Extra trees	8.18	15.3	0.428
Solid	SVR	9.12	16.31	0.35
Solid	Ridge	13.44	16.63	0.324
Solid	k-NN	9.95	16.74	0.316
Solid	GPBoost	19.14	19.85	0.038
Solid	MLP	8.03	20.35	-0.012
Solid	Mean	20.25	20.63	-0.039

Table S3. Per-study RMSE of each learner vs the best learner under grouped CV (Wilcoxon, Holm).

target	features	learner	reference	n	median_diff	ci_low	ci_high	rank_biserial	p	p_holm
biooil	base	cat	et	122	-0.2216	-0.8093	0.7455	0.0585	0.5749	1.0
biooil	base	gpboost	et	122	0.9174	0.1383	2.3352	0.2656	0.0109	0.0973
biooil	base	hgb	et	122	0.3776	-0.319	1.0831	0.098	0.3478	1.0
biooil	base	knn	et	122	0.188	-0.3295	0.7372	0.0649	0.5338	1.0
biooil	base	lgbm	et	122	-0.36	-1.0746	0.441	-0.0801	0.4426	1.0
biooil	base	mean	et	122	1.1666	0.324	2.7974	0.3197	0.0022	0.0218
biooil	base	mlp	et	122	1.4015	0.8269	2.3835	0.4064	0.0001	0.0011
biooil	base	rf	et	122	-0.1996	-0.9058	0.2122	-0.0641	0.5389	1.0
biooil	base	ridge	et	122	0.7065	-0.0266	1.4461	0.2659	0.0108	0.0973
biooil	base	svr	et	122	0.2585	-0.1276	1.0054	0.0964	0.3557	1.0
biooil	base	xgb	et	122	-0.6037	-1.2874	0.4911	-0.135	0.1956	1.0
biooil	base+P	cat	lgbm	122	0.4598	-0.0338	1.0689	0.2486	0.0172	0.1204
biooil	base+P	et	lgbm	122	0.6415	-0.0084	1.3662	0.1651	0.1135	0.3404
biooil	base+P	gpboost	lgbm	122	1.5691	0.4798	2.3756	0.3024	0.0037	0.03
biooil	base+P	hgb	lgbm	122	0.3768	-0.0465	0.8419	0.2104	0.0437	0.2184
biooil	base+P	knn	lgbm	122	1.0143	-0.0005	2.0762	0.2219	0.0334	0.2005
biooil	base+P	mean	lgbm	122	1.8974	0.2815	2.9184	0.3261	0.0018	0.0159
biooil	base+P	mlp	lgbm	122	3.008	2.1817	4.5248	0.5223	0.0	0.0
biooil	base+P	rf	lgbm	122	-0.1483	-0.4438	0.6064	0.0566	0.5872	0.6294
biooil	base+P	ridge	lgbm	122	1.9552	0.7435	2.5843	0.3347	0.0013	0.0134

target	features	learner	reference	n	median_diff	ci_low	ci_high	rank_biserial	p	p_holm
biooil	base+P	svr	lgbm	122	1.0249	-0.2102	1.5866	0.1889	0.0703	0.2811
biooil	base+P	xgb	lgbm	122	0.1263	-0.1065	0.5132	0.1049	0.3147	0.6294
solid	base	cat	lgbm	120	0.2643	-0.0284	0.5413	0.1388	0.1869	0.3737
solid	base	et	lgbm	120	0.5281	0.1758	1.169	0.2642	0.012	0.0601
solid	base	gpboost	lgbm	120	4.5226	3.3612	6.0919	0.7085	0.0	0.0
solid	base	hgb	lgbm	120	0.5105	-0.0365	0.7768	0.1994	0.058	0.2318
solid	base	knn	lgbm	120	1.3354	0.2329	2.1963	0.3204	0.0023	0.0139
solid	base	mean	lgbm	120	5.4683	4.3418	6.7941	0.7358	0.0	0.0
solid	base	mlp	lgbm	120	2.007	1.2778	3.6872	0.614	0.0	0.0
solid	base	rf	lgbm	120	0.234	-0.1967	0.5777	0.0829	0.4305	0.4305
solid	base	ridge	lgbm	120	1.6468	0.8181	2.6226	0.4556	0.0	0.0001
solid	base	svr	lgbm	120	0.9475	0.1213	1.7558	0.3295	0.0017	0.0121
solid	base	xgb	lgbm	120	0.3587	0.0882	0.5465	0.1964	0.0619	0.2318
solid	base+P	cat	lgbm	120	0.3388	-0.0327	0.9523	0.2011	0.0559	0.1677
solid	base+P	et	lgbm	120	0.9511	0.0248	1.5519	0.3215	0.0022	0.009
solid	base+P	gpboost	lgbm	120	4.9142	3.3016	6.3658	0.6986	0.0	0.0
solid	base+P	hgb	lgbm	120	0.2709	0.0949	0.5406	0.3397	0.0012	0.0062
solid	base+P	knn	lgbm	120	1.8679	1.0058	3.4828	0.4909	0.0	0.0
solid	base+P	mean	lgbm	120	5.5907	4.1452	6.9488	0.7331	0.0	0.0
solid	base+P	mlp	lgbm	120	3.5439	2.9145	4.2279	0.7449	0.0	0.0
solid	base+P	rf	lgbm	120	0.3941	-0.2203	0.8728	0.1598	0.1288	0.2576
solid	base+P	ridge	lgbm	120	1.7981	0.9084	2.6774	0.5245	0.0	0.0
solid	base+P	svr	lgbm	120	1.7353	0.6701	2.2464	0.4623	0.0	0.0001
solid	base+P	xgb	lgbm	120	0.0466	-0.2548	0.2825	0.0843	0.4229	0.4229

Table S4. Coverage for all α levels and settings.

Target	Setting	α	Calibration	Coverage (rows)	Coverage (studies)	Width (wt%)	Studies <0.8 coverage
Bio-oil	Development CV (unseen studies)	0.05	RC	0.758	0.73	33.1	0.42
Bio-oil	Development CV (unseen studies)	0.1	RC	0.612	0.57	22.8	0.66
Bio-oil	Development CV (unseen studies)	0.2	RC	0.442	0.398	14.5	0.85
Bio-oil	Development CV (unseen studies)	0.05	SC	0.965	0.953	76.4	0.08
Bio-oil	Development CV (unseen studies)	0.1	SC	0.93	0.909	59.1	0.16
Bio-oil	Development CV (unseen studies)	0.2	SC	0.852	0.822	42.9	0.26
Solid	Development CV (unseen studies)	0.05	RC	0.748	0.752	29.0	0.48
Solid	Development CV (unseen studies)	0.1	RC	0.615	0.614	20.6	0.66

Target	Setting	α	Calibration	Coverage (rows)	Coverage (studies)	Width (wt%)	Studies <0.8 coverage
Solid	Development CV (unseen studies)	0.2	RC	0.461	0.454	13.4	0.82
Solid	Development CV (unseen studies)	0.05	SC	0.95	0.955	60.1	0.08
Solid	Development CV (unseen studies)	0.1	SC	0.901	0.908	46.7	0.16
Solid	Development CV (unseen studies)	0.2	SC	0.807	0.815	33.4	0.35
Bio-oil	Locked test	0.05	RC	0.678	0.726	33.5	0.43
Bio-oil	Locked test	0.05	SC	0.874	0.95	68.8	0.11
Bio-oil	Locked test	0.1	RC	0.565	0.603	22.7	0.61
Bio-oil	Locked test	0.1	SC	0.82	0.895	54.8	0.21
Bio-oil	Locked test	0.2	RC	0.426	0.428	14.8	0.93
Bio-oil	Locked test	0.2	SC	0.748	0.801	39.7	0.32
Solid	Locked test	0.05	RC	0.591	0.582	28.5	0.63
Solid	Locked test	0.05	SC	0.908	0.882	59.0	0.17
Solid	Locked test	0.1	RC	0.44	0.457	21.0	0.77
Solid	Locked test	0.1	SC	0.864	0.821	46.7	0.23
Solid	Locked test	0.2	RC	0.289	0.292	13.3	0.93
Solid	Locked test	0.2	SC	0.707	0.677	35.4	0.57
Bio-oil	Leave-family-out (mean)	0.05	RC	0.738	0.722	32.4	0.44
Bio-oil	Leave-family-out (mean)	0.05	SC	0.93	0.942	74.1	0.09
Bio-oil	Leave-family-out (mean)	0.1	RC	0.569	0.554	22.5	0.62
Bio-oil	Leave-family-out (mean)	0.1	SC	0.888	0.894	57.6	0.16
Bio-oil	Leave-family-out (mean)	0.2	RC	0.335	0.329	14.5	0.83
Bio-oil	Leave-family-out (mean)	0.2	SC	0.815	0.804	41.8	0.29
Bio-oil	Temporal (≤2018 → >2018)	0.05	RC	0.574	0.55	29.6	0.69
Bio-oil	Temporal (≤2018 → >2018)	0.05	SC	0.885	0.906	75.9	0.17
Bio-oil	Temporal (≤2018 → >2018)	0.1	RC	0.456	0.422	21.9	0.84
Bio-oil	Temporal (≤2018 → >2018)	0.1	SC	0.846	0.853	64.1	0.28
Bio-oil	Temporal (≤2018 → >2018)	0.2	RC	0.306	0.286	13.5	0.97

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Target	Setting	α	Calibration	Coverage (rows)	Coverage (studies)	Width (wt%)	Studies <0.8 coverage
Bio-oil	Temporal (≤2018 → >2018)	0.2	SC	0.748	0.743	46.1	0.41
Solid	Leave-family-out (mean)	0.05	RC	0.683	0.744	28.7	0.37
Solid	Leave-family-out (mean)	0.05	SC	0.89	0.926	58.9	0.12
Solid	Leave-family-out (mean)	0.1	RC	0.548	0.596	20.5	0.57
Solid	Leave-family-out (mean)	0.1	SC	0.858	0.896	46.5	0.16
Solid	Leave-family-out (mean)	0.2	RC	0.35	0.369	13.3	0.87
Solid	Leave-family-out (mean)	0.2	SC	0.748	0.795	34.2	0.33
Solid	Temporal (≤2018 → >2018)	0.05	RC	0.648	0.692	25.7	0.53
Solid	Temporal (≤2018 → >2018)	0.05	SC	0.909	0.925	58.6	0.14
Solid	Temporal (≤2018 → >2018)	0.1	RC	0.521	0.559	18.7	0.72
Solid	Temporal (≤2018 → >2018)	0.1	SC	0.877	0.897	49.0	0.14
Solid	Temporal (≤2018 → >2018)	0.2	RC	0.384	0.418	12.7	0.88
Solid	Temporal (≤2018 → >2018)	0.2	SC	0.809	0.832	39.4	0.27
Solid	External 4TU	0.05	RC	0.681	0.724	28.7	0.49
Solid	External 4TU	0.05	SC	0.964	0.948	55.4	0.12
Solid	External 4TU	0.1	RC	0.534	0.549	20.7	0.7
Solid	External 4TU	0.1	SC	0.934	0.918	47.3	0.16
Solid	External 4TU	0.2	RC	0.404	0.416	13.2	0.83
Solid	External 4TU	0.2	SC	0.783	0.809	34.2	0.32

Table S5. Sensitivity of coverage and accuracy to α and to the number of outer folds (seed 11).

target	features	outer_folds	alpha	calibration	Coverage (studies)	Width (wt%)	RMSE
Bio-oil	base	5.0	0.05	RC	0.704	32.054	15.609
Bio-oil	base	5.0	0.05	SC	0.962	77.906	15.609
Bio-oil	base	5.0	0.1	RC	0.556	22.655	15.609
Bio-oil	base	5.0	0.1	SC	0.921	58.656	15.609
Bio-oil	base	5.0	0.2	RC	0.396	14.545	15.609
Bio-oil	base	5.0	0.2	SC	0.837	43.221	15.609
Bio-oil	base	10.0	0.05	RC	0.702	31.687	15.588
Bio-oil	base	10.0	0.05	SC	0.948	69.763	15.588
Bio-oil	base	10.0	0.1	RC	0.547	22.587	15.588
Bio-oil	base	10.0	0.1	SC	0.902	54.794	15.588
Bio-oil	base	10.0	0.2	RC	0.406	14.899	15.588
Bio-oil	base	10.0	0.2	SC	0.813	40.993	15.588
Bio-oil	base+P	5.0	0.05	RC	0.689	32.286	16.422
Bio-oil	base+P	5.0	0.05	SC	0.954	79.124	16.422
Bio-oil	base+P	5.0	0.1	RC	0.545	22.851	16.422
Bio-oil	base+P	5.0	0.1	SC	0.909	60.165	16.422
Bio-oil	base+P	5.0	0.2	RC	0.369	14.546	16.422
Bio-oil	base+P	5.0	0.2	SC	0.826	44.653	16.422
Bio-oil	base+P	10.0	0.05	RC	0.714	32.104	15.654
Bio-oil	base+P	10.0	0.05	SC	0.951	74.833	15.654
Bio-oil	base+P	10.0	0.1	RC	0.556	22.555	15.654
Bio-oil	base+P	10.0	0.1	SC	0.904	58.574	15.654
Bio-oil	base+P	10.0	0.2	RC	0.394	14.439	15.654
Bio-oil	base+P	10.0	0.2	SC	0.808	42.157	15.654
Solid	base	5.0	0.05	RC	0.722	28.444	14.043
Solid	base	5.0	0.05	SC	0.962	62.917	14.043
Solid	base	5.0	0.1	RC	0.588	20.277	14.043
Solid	base	5.0	0.1	SC	0.914	48.778	14.043
Solid	base	5.0	0.2	RC	0.433	13.061	14.043
Solid	base	5.0	0.2	SC	0.813	35.711	14.043
Solid	base	10.0	0.05	RC	0.747	28.414	14.165
Solid	base	10.0	0.05	SC	0.961	61.863	14.165
Solid	base	10.0	0.1	RC	0.6	20.358	14.165
Solid	base	10.0	0.1	SC	0.902	47.455	14.165
Solid	base	10.0	0.2	RC	0.438	13.135	14.165
Solid	base	10.0	0.2	SC	0.817	34.51	14.165
Solid	base+P	5.0	0.05	RC	0.729	28.583	14.107
Solid	base+P	5.0	0.05	SC	0.961	63.523	14.107
Solid	base+P	5.0	0.1	RC	0.589	20.371	14.107
Solid	base+P	5.0	0.1	SC	0.915	49.307	14.107
Solid	base+P	5.0	0.2	RC	0.438	13.383	14.107
Solid	base+P	5.0	0.2	SC	0.815	34.849	14.107
Solid	base+P	10.0	0.05	RC	0.74	28.562	13.938
Solid	base+P	10.0	0.05	SC	0.959	60.668	13.938
Solid	base+P	10.0	0.1	RC	0.602	20.562	13.938
Solid	base+P	10.0	0.1	SC	0.913	46.926	13.938
Solid	base+P	10.0	0.2	RC	0.439	13.415	13.938
Solid	base+P	10.0	0.2	SC	0.804	33.714	13.938

Table S6. k-shot results with method-of-moments vs REML estimates of the between-study variance.

target	features	estimator	k	Median study RMSE	Median study coverage	Width (wt%)
Bio-oil	base	MoM	0.0	12.343	1.0	55.749
Bio-oil	base	MoM	1.0	9.447	0.978	40.956
Bio-oil	base	MoM	2.0	8.691	0.97	37.729
Bio-oil	base	MoM	3.0	8.523	0.967	36.551
Bio-oil	base	MoM	5.0	8.015	0.98	35.879
Bio-oil	base	MoM	10.0	7.847	0.972	33.629
Bio-oil	base	REML	0.0	12.343	1.0	55.749
Bio-oil	base	REML	1.0	9.335	0.97	40.504
Bio-oil	base	REML	2.0	8.718	0.971	37.64
Bio-oil	base	REML	3.0	8.474	0.965	36.256
Bio-oil	base	REML	5.0	8.005	0.973	35.931
Bio-oil	base	REML	10.0	7.845	0.967	33.755
Bio-oil	base+P	MoM	0.0	12.613	1.0	58.715
Bio-oil	base+P	MoM	1.0	10.057	0.98	42.279
Bio-oil	base+P	MoM	2.0	8.838	0.979	38.928
Bio-oil	base+P	MoM	3.0	8.4	0.973	37.576
Bio-oil	base+P	MoM	5.0	8.065	0.983	36.164
Bio-oil	base+P	MoM	10.0	7.59	0.975	33.853
Bio-oil	base+P	REML	0.0	12.613	1.0	58.715
Bio-oil	base+P	REML	1.0	9.371	0.98	42.301
Bio-oil	base+P	REML	2.0	8.812	0.974	38.887
Bio-oil	base+P	REML	3.0	8.416	0.965	37.413
Bio-oil	base+P	REML	5.0	8.073	0.981	36.142
Bio-oil	base+P	REML	10.0	7.584	0.975	33.958
Solid	base	MoM	0.0	10.537	1.0	46.419
Solid	base	MoM	1.0	8.876	0.992	39.127
Solid	base	MoM	2.0	8.268	0.996	35.71
Solid	base	MoM	3.0	7.749	0.995	35.146
Solid	base	MoM	5.0	7.136	0.995	33.432
Solid	base	MoM	10.0	6.721	1.0	34.108
Solid	base	REML	0.0	10.537	1.0	46.419
Solid	base	REML	1.0	9.192	0.998	44.343
Solid	base	REML	2.0	8.612	0.997	39.718
Solid	base	REML	3.0	8.594	0.997	38.939
Solid	base	REML	5.0	8.312	0.999	36.034
Solid	base	REML	10.0	7.747	1.0	35.651
Solid	base+P	MoM	0.0	10.674	1.0	46.636
Solid	base+P	MoM	1.0	8.89	0.994	39.247
Solid	base+P	MoM	2.0	8.161	0.991	35.68
Solid	base+P	MoM	3.0	7.876	0.992	34.247
Solid	base+P	MoM	5.0	7.742	0.992	32.88
Solid	base+P	MoM	10.0	6.933	0.999	33.515
Solid	base+P	REML	0.0	10.674	1.0	46.636

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target	features	estimator	k	Median study RMSE	Median study coverage	Width (wt%)
Solid	base+P	REML	1.0	9.335	1.0	42.411
Solid	base+P	REML	2.0	8.457	0.999	39.1
Solid	base+P	REML	3.0	8.424	0.999	37.731
Solid	base+P	REML	5.0	8.294	1.0	35.803
Solid	base+P	REML	10.0	8.064	1.0	35.138

Table S7. Study-grouped CV of the STRATA base learner with tuned vs library-default hyper-parameters.

target	features	config	rmse	r2	median_study_rmse
Bio-oil	base	tuned	15.848	0.226	12.729
Bio-oil	base	default	16.046	0.206	12.418
Bio-oil	base+P	tuned	16.181	0.193	12.244
Bio-oil	base+P	default	17.206	0.087	13.555
Solid	base	tuned	14.076	0.516	10.835
Solid	base	default	14.035	0.519	10.078
Solid	base+P	tuned	13.818	0.534	10.913
Solid	base+P	default	14.821	0.463	11.439

S4. Exploratory: locally adaptive study-calibrated intervals (post hoc)

Motivation: constant-width study-calibrated intervals are wide. Hypothesis: out-of-study errors are heteroscedastic, so dividing conformity scores by a learned difficulty $\hat{\sigma}(x)$ narrows intervals for easy regimes without losing study-level coverage. A shallow gradient-boosting model of $\log(|r|+0.5)$ was cross-fitted by study on the out-of-study residuals of the training studies; scores $|r|/\hat{\sigma}$ were calibrated with the same study-weighted quantile, and intervals were $f(x)\pm\hat{q}\,\hat{\sigma}(x)$. For bio-oil, SC-N gave study-weighted coverage 0.91 (SC 0.91) with median width 55.6 vs 59.5 wt% and mean interval score 84.2 vs 77.3; for solids, coverage 0.92 vs 0.91, median width 53.3 vs 45.7 wt%, interval score 69.5 vs 65.2. The adaptive variant therefore did not improve interval efficiency — both widths and interval scores were worse — although coverage across process subtypes became more uniform (largest deviation

from 0.90: 0.03 vs 0.07 for solids). This negative result is consistent with the main finding that the dominant error component is a study offset that input features cannot localise. Because this variant was added after inspecting pipeline-verification results, it is reported as exploratory and is not part of the main claims.

Table S8. Exploratory (post hoc): constant-width vs locally adaptive study-calibrated intervals (α = 0.1, base+P).

target	method	coverage_rows	coverage_studies	mean_width	median_width	winkler	coverage_by_process	max_abs_process_dev
Bio-oil	RC	0.612	0.57	22.832	22.756	107.655	{'HTL': 0.609, 'HTL_solvothermal': 0.52}	0.38
Bio-oil	SC	0.93	0.909	59.079	59.499	77.294	{'HTL': 0.932, 'HTL_solvothermal': 0.881}	0.032
Bio-oil	SC-N	0.915	0.914	63.008	55.638	84.231	{'HTL': 0.929, 'HTL_solvothermal': 0.897}	0.029
Solid	RC	0.615	0.614	20.564	20.554	93.706	{'HTC': 0.527, 'HTL': 0.644, 'HTL_solvothermal': 0.577}	0.373
Solid	SC	0.901	0.908	46.672	45.651	65.196	{'HTC': 0.827, 'HTL': 0.916, 'HTL_solvothermal': 0.905}	0.073
Solid	SC-N	0.921	0.92	57.302	53.283	69.497	{'HTC': 0.868, 'HTL': 0.925, 'HTL_solvothermal': 0.92}	0.032

S5. Error analysis by subgroup

Table S9. Out-of-study errors by subgroup (seed-averaged outer-fold predictions, base features).

target	dimension	level	n	n_studies	rmse	bias	mae
Bio-oil	family	Agricultural Residues	804	56	14.0	-1.21	10.47
Bio-oil	family	Animal Manure	36	4	14.7	-11.29	11.63
Bio-oil	family	Aquatic Biomass	19	3	20.23	-9.17	19.07
Bio-oil	family	Herbaceous Biomass	4	1	20.98	-14.32	17.42
Bio-oil	family	Lignin-rich Streams (Technical)	189	17	17.52	0.24	13.88
Bio-oil	family	Mixed Biomass	205	16	16.63	-2.93	12.93
Bio-oil	family	Other	91	6	33.89	14.33	25.89
Bio-oil	family	Plastics / Polymer Blends	11	2	33.62	-31.14	31.14
Bio-oil	family	Waste Biomass / Sludge	40	6	12.78	-6.31	9.65
Bio-oil	family	Waste Wood / Construction	33	5	19.41	0.27	16.48
Bio-oil	family	Woody Biomass / Hardwood	417	27	12.43	-0.38	8.58
Bio-oil	family	Woody Biomass / Softwood	352	29	10.72	-4.15	7.94
Bio-oil	family	Woody Biomass / Unspecified	60	9	16.24	5.1	11.97
Bio-oil	process	HTL	1513	91	13.85	-1.99	10.09
Bio-oil	process	HTL_solvothermal	748	46	18.5	0.33	13.36
Bio-oil	extraction_solvent	acetone	783	47	12.94	-2.07	9.61
Bio-oil	extraction_solvent	alcohol	191	14	13.92	-0.41	12.32
Bio-oil	extraction_solvent	chlorinated_other	2	1	10.22	9.25	9.25
Bio-oil	extraction_solvent	dcm	365	24	16.47	-3.35	12.63
Bio-oil	extraction_solvent	diethyl_ether	46	6	14.21	-9.52	13.4
Bio-oil	extraction_solvent	ethyl_acetate	179	10	14.67	-6.55	12.03
Bio-oil	extraction_solvent	hydrocarbon	149	6	10.81	0.32	8.13
Bio-oil	extraction_solvent	none_named	514	31	19.8	2.7	12.34
Bio-oil	extraction_solvent	thf	32	4	19.29	9.71	13.66
Bio-oil	temperature_bin	250-300	1129	95	13.35	-1.7	9.96
Bio-oil	temperature_bin	300-350	565	76	15.7	1.0	11.15
Bio-oil	temperature_	<=250	354	50	21.12	-1.13	15.01

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target	dimension	level	n	n_studies	rmse	bias	mae
	bin						
Bio-oil	temperature_bin	>350	213	36	14.96	-4.76	11.27
Bio-oil	catalyst_present	no	1061	104	15.02	-1.97	11.11
Bio-oil	catalyst_present	yes	1200	91	15.99	-0.56	11.23
Bio-oil	yield_basis_daf	daf	332	17	13.35	-0.48	9.27
Bio-oil	yield_basis_daf	dry/other	1929	106	15.89	-1.35	11.5
Solid	family	Agricultural Residues	781	55	12.45	1.7	9.73
Solid	family	Animal Manure	33	3	10.25	-4.69	9.06
Solid	family	Aquatic Biomass	19	3	17.15	-12.64	15.62
Solid	family	Herbaceous Biomass	44	4	18.34	-11.52	13.57
Solid	family	Lignin-rich Streams (Technical)	221	21	14.86	-6.18	11.12
Solid	family	Mixed Biomass	196	15	12.99	2.26	10.27
Solid	family	Model Compound	5	1	4.69	-1.35	4.18
Solid	family	Other	67	7	31.91	-3.95	27.2
Solid	family	Waste Biomass / Sludge	72	5	19.44	8.84	16.81
Solid	family	Waste Wood / Construction	31	4	7.36	1.37	6.0
Solid	family	Woody Biomass / Hardwood	304	23	12.36	-1.21	8.98
Solid	family	Woody Biomass / Softwood	350	23	10.21	2.08	7.34
Solid	family	Woody Biomass / Unspecified	33	7	11.07	-1.85	8.32
Solid	process	HTC	380	16	14.93	-0.99	11.49
Solid	process	HTL	1180	80	12.66	-1.05	9.0
Solid	process	HTL_solvothermal	596	40	15.16	3.07	11.9
Solid	extraction_solvent	acetone	690	42	13.81	1.63	9.52
Solid	extraction_solvent	alcohol	153	13	10.8	-0.99	9.05
Solid	extraction_solvent	dcm	306	21	13.02	-0.92	9.87
Solid	extraction_solvent	diethyl_ether	39	3	13.95	-5.95	11.64
Solid	extraction_solvent	ethyl_acetate	111	8	17.08	-4.3	14.0
Solid	extraction_solvent	hydrocarbon	80	2	17.52	14.17	16.13

target	dimension	level	n	n_studies	rmse	bias	mae
Solid	extraction_solvent	none_named	745	37	13.86	-1.14	10.24
Solid	extraction_solvent	thf	32	4	7.83	-1.59	5.28
Solid	temperature_bin	250-300	962	89	13.74	1.03	10.4
Solid	temperature_bin	300-350	458	70	12.66	-1.19	8.8
Solid	temperature_bin	<=250	546	61	14.88	-0.24	11.24
Solid	temperature_bin	>350	190	29	13.57	-0.56	10.04
Solid	catalyst_present	no	1291	104	13.43	-0.01	10.23
Solid	catalyst_present	yes	865	81	14.34	0.25	10.26
Solid	yield_basis_daf	daf	418	16	14.31	1.08	11.28
Solid	yield_basis_daf	dry/other	1738	105	13.68	-0.14	9.99

S6. STRATA with different base learners (E9)

Table S10. STRATA with different base learners (development study-folds, seed 11, base + P features, α = 0.1). The mean predictor is the ‘no-ML’ reference.

Target	Base learner	Coverage RC	Coverage SC	Width RC (wt%)	Width SC (wt%)	Pooled RMSE	Median study RMSE k=0	Median study RMSE k=3 (EB)
Bio-oil	Extra trees	0.575	0.903	25.1	54.062	16.264	13.175	8.216
Bio-oil	k-NN	0.647	0.91	30.43	58.999	16.935	13.392	8.831
Bio-oil	LightGBM	0.553	0.904	22.542	53.69	15.869	12.512	8.415
Bio-oil	Mean	0.86	0.901	54.59	63.046	18.078	13.355	9.15
Bio-oil	Random forest	0.588	0.904	24.544	53.502	16.09	12.286	8.183
Bio-oil	Ridge	0.8	0.899	43.064	58.986	17.523	14.159	8.727
Bio-oil	XGBoost	0.559	0.904	22.495	58.394	16.181	12.244	8.4
Solid	Extra trees	0.672	0.899	25.849	49.029	15.237	11.016	8.545
Solid	k-NN	0.727	0.909	33.243	50.757	16.438	13.815	10.325
Solid	LightGBM	0.605	0.913	20.15	45.962	13.837	10.548	7.849
Solid	Mean	0.935	0.904	63.637	59.37	20.599	16.982	9.889
Solid	Random forest	0.7	0.919	26.475	47.45	14.158	10.865	7.856
Solid	Ridge	0.846	0.893	43.308	50.622	16.973	13.035	9.19
Solid	XGBoost	0.602	0.913	20.768	46.847	13.818	10.913	7.876

S7. Reproducibility

`python run_all.py --mode full` regenerates every result file, table and figure; `python manuscript/src/build_manuscript.py` regenerates this document and the manuscript from those files. Environment details (library versions, CPU) are written to `results/experiment_logs/environment.json` at run time.

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